PROJECT REPORT

On

ADVANCING ACUTE RESPIRATORY INFECTION DIAGNOSIS THROUGH MULTIMODAL VISION TRANSFORMER FOR RESPIRATORY DISEASE CLASSIFICATION

A Project Report submitted to

Alliance University, Bengaluru

For the award of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

by

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COMPUTER SCIENCE AND ENGINEERING
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CERTIFICATE

This is to certify that the project work entitled **"Advancing Acute Respiratory Infection Diagnosis through Vision Transformer Integration with Multimodal Medical Imaging"** submitted by **K. Nithin Kumar [2022BCSE07AED659]**, **G. Sai Smaran [2022BCSE07AED573]**, **N. Girish [2022BCSE07AED257]**, and **R. Guna Vardhan [2022BCSE07AED099]** in partial fulfillment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** of Alliance University, is a bonafide work carried out under our supervision and guidance during the academic year **2025–2026**.This thesis report embodies the result of original work and studies conducted by the students and the contents do not form the basis for the award of any other degree to the candidate or anybody else.

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DECLARATION

We hereby declare that the project entitled "**Advancing Acute Respiratory Infection Diagnosis through Vision Transformer Integration with Multimodal Medical Imaging**" submitted by us in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** of ASAC, Alliance University, is a record of our work carried out under the supervision and guidance of **A. Ezil Sem Leni, Professor, Department of Computer Science and Engineering**.

We confirm that this report truly represents the work undertaken as a part of our project work. This work is not a replication of work done previously by any other person. We also confirm that the contents of the report and the views contained therein have been discussed and deliberated with the faculty guide.

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PREFACE

The rapid advancement of Artificial Intelligence and Deep Learning technologies has significantly transformed the field of healthcare, particularly in medical imaging and disease diagnosis. This project, titled **"Advancing Acute Respiratory Infection Diagnosis through Vision Transformer Integration with Multimodal Medical Imaging,"** explores the application of modern transformer-based architectures in improving diagnostic accuracy for respiratory diseases.

The primary objective of this project is to develop an intelligent system capable of analyzing multimodal medical data, including X-ray images, CT scans, and clinical inputs, to assist in the early and accurate detection of Acute Respiratory Infections (ARIs). By leveraging the capabilities of Vision Transformers along with traditional deep learning techniques, this work aims to address the limitations of conventional diagnostic approaches.

This project has provided us with an opportunity to gain practical knowledge in machine learning, deep learning, medical image processing, and model evaluation techniques. It also enhanced our understanding of real-world healthcare challenges and the importance of integrating advanced computational methods to support medical decision-making.

We hope that the outcomes of this project contribute towards the development of efficient and reliable diagnostic tools in the healthcare domain and serve as a foundation for further research in this area.

ABSTRACT

Acute respiratory diseases, lung diseases such as COVID-19, pneumonia, and tuberculosis continue to represent significant challenges to world healthcare systems due to their huge prevalence and mortality rates. Accurate diagnosis at early stages of these conditions is essential for effective treatment and improved outcomes. Diagnosis imaging methods such as chest X-rays and computed tomography (CT) scans play a crucial role in detecting respiratory abnormalities; however, manual interpretation of these images is time-consuming and requires expert radiologists, often leading to delays and variability in diagnosis.

This project presents an advanced deep learning-based approach for the classification of respiratory diseases using a **Multimodal Vision Transformer (ViT)** framework. The proposed system integrates information from both CT scan images and chest X-ray images to leverage complementary diagnostic features from multiple imaging modalities. A dual-branch transformer architecture is employed, where each modality is processed independently to extract meaningful features, which are then fused for final classification.

The dataset used consists of publicly available CT and chest X-ray images categorized into four classes: COVID-19, Pneumonia, Tuberculosis, and Normal. All images are preprocessed and standardized to a resolution of 224×224 pixels to ensure consistency in model training. The Vision Transformer model utilizes self-attention mechanisms to capture global dependencies within images, enabling improved feature extraction compared to traditional Convolutional Neural Networks (CNNs).

Experimental results demonstrate that the proposed multimodal approach achieves a validation accuracy of approximately **88%**, with strong performance across evaluation metrics such as precision, recall, and F1-score. The results indicate that combining multiple imaging modalities significantly enhances classification performance compared to single-modality approaches.

In conclusion, the proposed system provides an efficient and reliable solution for automated respiratory disease diagnosis. It has the potential to assist healthcare professionals in clinical decision-making, reduce diagnostic workload, and improve screening efficiency. This work also highlights the importance of transformer-based architectures in advancing medical image analysis and paves the way for future research in multimodal healthcare AI systems.

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LIST OF ABBREVIATIONS & SYMBOLS

Abbreviation / Symbol	**Meaning**
Adam	Adaptive Moment Estimation (Optimizer)
ARI	Acute Respiratory Infection
AUC	Area Under the Curve
CNN	Convolutional Neural Network
CT	Computed Tomography
CXR	Chest X-Ray
d_k	Key Dimension in Scaled Dot-Product Attention
F1	F1 Score (Harmonic Mean of Precision and Recall)
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
MHA	Multi-Head Attention
N	Number of Image Patches
P	Patch Size
Q, K, V	Query, Key, Value matrices in Self-Attention
ROC	Receiver Operating Characteristic
TB	Tuberculosis
ViT	Vision Transformer
α (Alpha)	Learning Rate
Σ (Sigma)	Summation

CHAPTER 1
INTRODUCTION

1.1 Background

Respiratory diseases remain one of the most critical global health concerns, contributing significantly to mortality and morbidity rates across the world. Conditions such as COVID-19, pneumonia, and tuberculosis affect millions of individuals annually and place immense pressure on healthcare systems. Early and accurate diagnosis of these diseases is essential to ensure timely treatment, reduce complications, and improve patient survival rates. However, conventional diagnostic procedures often rely on direct inspection of medical images and clinical symptoms, which can be protracted and fallible to human mistakes.

Radiological diagnostics such as chest X-rays and computed tomography (CT) scans play a vital role in diagnosing respiratory conditions. Chest X-rays are widely used due to their accessibility and low cost, whereas CT scans provide detailed three-dimensional views of lung structures, enabling better identification of abnormalities. Despite their advantages, interpreting these images requires specialized expertise and experience. In situations where there is a shortage of skilled radiologists or during large-scale outbreaks, the diagnostic process becomes slower and less efficient.

Recent advancements in Artificial Intelligence (AI) and Deep Learning (DL) have introduced new possibilities for automating medical image analysis. These technologies have showed striking performance in detecting patterns and features that may not be easily visible to the human eye. Among different deep learning techniques, Convolutional Neural Networks (CNNs) have been widely used for image arrangement tasks, including medical image analysis. CNN-based models have shown hopeful results in identifying diseases from chest X-ray images. However, they central focus on local attributes and may struggle to capture extented dependencies within complex medical images.

To overcome these limitations, attention mechanism architectures, particularly Vision Transformers (ViTs), have gained considerable focus in recent years. Unlike CNNs, Vision Transformers use recursive attention mechanisms to capture universal relations between different zones of an image. This allows them to process images more efficiently and significant features from complex patterns. Outermost, the integration of several data sources, known as diversified learning approach by multimodals, has emerged as a powerful approach in clinical evaluation. By integrating data from different imaging modalities, such as CT scans and X-rays, it is feasible more accurate and reliable results.

This project looks into the utilization of Vision Transformers in combination with multimodal medical imaging to enhance the analysis of acute respiratory infections. By capitalizing sophisticated layered learning methods and integrating diverse data sources, the proposed system aims to enhance diagnostic accuracy and support healthcare professionals in decision-making processes.

1.1.1 Domain Overview

The specialty of this project lies at the convergence of Artificial Intelligence, Deep Learning, and healthcare data image analysis. Diagnostic visualization review is a rapidly advancing field that focuses on withdrawing meaningful information from medical images to aid in diagnosis, medical management design, and disease monitoring. Alongside, increasing availability of computerized medical records, there is a mounting requirements for computerized processes that can efficiently analyze massive quantites of medical images.

Artificial Intelligence has transformed the healthcare sector by enabling the development of intelligent systems capable of learning from data and making predictions. In particular, deep learning techniques have shown exceptional performance in image recognition and classification tasks. These models can automatically learn sophisticated characteristics from raw data, eliminating the need for manual feature extraction.

Alongside this domain, respiratory disease detection using imaging techniques has acquired significant importance. Chest X-rays and CT scans are among the most commonly used imaging modalities for diagnosing lung-related conditions. Each modality provides unique information: X-rays offer a quick summary of lung conditions, while CT scans provide detailed insights into inner framework. Combining these modalities allows for a more comprehensive understanding of the disease.

Vision Transformers shows a recent development in deep learning, inspired by transformer models originally conceived for natural language processing. These models treat images as series of patches and apply attention mechanisms to comprehend connections between different parts of the image. This approach allows better depiction of global features, making it highly suitable for complex medical imaging tasks.

Multimodal learning continues to enrich the capabilities of AI systems by incorporating information from multiple sources. With respect to this project, combining CT scan images and chest X-ray images enables the model to use complementary features from both approches. This leads to improved model precision and more accurate disease detection.

The integration of these leading edge technologies has the potential to revolutionize healthcare by providing faster, more precise, and scalable diagnostic solutions. It reduces reliance on manual analysis and enables healthcare professionals to focus on critical decision-making tasks.

1.2 Problem Statement

Acute Respiratory Infections (ARIs), including illnesses such as COVID-19, pneumonia, and tuberculosis, present important challenges in terms of diagnosis due to intersecting symptoms and difference in imaging design. Conventional diagnostic approaches mainly rely on manual reading of medical images, which is lengthy, prone to human error, and dependent on the skill of radiologists. In many cases, especially during massive outbreak or in settings, the availability of skilled professionals is insufficient to handle the volume of diagnostic cases.

Moreover, existing automated systems regular focus on a single imaging modality, such as either chest X-rays or CT scans, confining their ability to capture the complete scoop of diagnostic information. These systems might neglect to fully exploit the complementary nature of different imaging techniques, leading to reduced accuracy and reliability in disease classification. Additionally, conventional deep learning models like Convolutional Neural Networks (CNNs) are primarily designed to capture local features and may not effectively model long-range dependencies within complex medical images.

Therefore, there is a need for an leading diagnostic system that can surmount these restrictive integrating multiple imaging modalities and utilizing modern deep learning architectures. The challenge lies in developing a model that can efficiently handel heterogeneous medical data, extract meaningful features, and provide accurate classification of respiratory diseases.

This project addresses these challenges by proposing a Multimodal Vision Transformer-based approach for respiratory disease diagnosis. The system aims to combine CT scan and chest X-ray images, leverage self-attention mechanisms for global feature extraction, and improve overall diagnostic performance. By doing so, it seeks to provide a reliable, efficient, and scalable solution that can assist healthcare professionals in early disease detection and decision-making.

1.3 Objectives

- To develop an automated system for precise detection of respiratory diseases such as COVID-19, pneumonia, and tuberculosis using medical imaging.
- To use multimodal data by combining chest X-ray and CT scan images for improved diagnostic performance.
- To implement a Vision Transformer (ViT)-based model for noting down global features and enhancing classification accuracy.
- To overcome constrains of traditional Convolutional Neural Networks (CNNs) in handling complex medical images.
- To enhance early diagnosis and assist healthcare professionals in clinical decision-making.
- To reduce diagnosis time and reliances on manual interpretation of medical images.

1.4 Scope

The sim of this project focuses on developing an leading diagnostic system for respiratory diseases using multimodal medical imaging and deep learning techniques. It includes the use of chest X-ray and CT scan images to enhance disease classification accuracy by leveraging complementary information from both modalities.

The project primarily covers the design and execution of a Vision Transformer (ViT)-based model capable of capturing global image features through self focusing mechanisms. It aims to overcome the limitations of traditional Convolutional Neural Networks (CNNs) by effectively modeling complex patterns and long-range dependencies in medical images.

Additionally, the system is designed to help healthcare professionals by providing faster and more reliable predictions for diseases such as COVID-19, pneumonia, and tuberculosis. It helps early diagnosis, reduces manual workload, and improves decision-making in clinical settings.

However, the scope is limited to image based diagnosis and does not contain clinical data such as patient history or laboratory results. The system is designed as a helpful tool and not a replacement for medical experts.

CHAPTER 2

LITERATURE SURVEY

2.1 Existing System

The diagnosis of respiratory diseases has conventionally relied on clinical examination, laboratory tests, and medical imaging techniques such as chest X-rays and computed tomography (CT) scans. Over time, advancements in imaging technology have improved the ability of healthcare professionals to detect irregularities in lung structures. However, the explanation of these images continues to depend heavily on the skill of radiologists, making the process time-consuming and sometimes inconsistent.

In recent years, the unification of Artificial Intelligence (AI) into healthcare has led to the development of automated systems designed to help in medical image analysis. Among these, machine learning and deep learning models have been broadly researched for detecting and classifying respiratory diseases. These systems aim to reduce human effort, improve diagnostic speed, and enhance accuracy.

Early AI-based diagnostic systems mainly relied on traditional machine learning algorithms such as Support Vector Machines (SVM), Decision Trees, and Random Forests. These models required manual feature extraction, where domain experts had to identify relevant features from medical images. This process was not only labor intensive but also limited in its ability to capture complex patterns present in medical data.

The introduction of deep learning, particularly Convolutional Neural Networks (CNNs), marked a significant shift in medical image analysis. CNNs are capable of automatically learning hierarchical features from raw image data, eliminating the need for manual feature engineering. As a result, CNN-based models have been widely adopted for tasks such as image classification, segmentation, and disease detection. In the context of respiratory diseases, CNNs have demonstrated promising results in identifying conditions such as pneumonia and COVID-19 from chest X-ray images.

Despite their success, existing systems still face several challenges. Most models are designed to work with a single type of input data, such as only chest X-rays or only CT scans. This limits their ability to leverage complementary information from different imaging modalities. Additionally, CNNs are inherently designed to focus on local features using convolutional filters, which may restrict their ability to capture global relationships across the entire image. This limitation can affect performance when dealing with complex medical images that require understanding of spatial dependencies.

Furthermore, many existing systems lack robustness when applied to diverse datasets. Variations in image quality, resolution, and patient demographics can impact model performance. In real-world clinical settings, data heterogeneity is a major concern, and systems must be capable of generalizing across different conditions.

Another limitation of existing systems is the lack of interpretability. While deep learning models can achieve high accuracy, they often function as “black boxes,” making it difficult for healthcare professionals to understand how decisions are made. This lack of transparency can reduce trust in automated systems, particularly in critical applications such as disease diagnosis.

Overall, while existing AI-based systems have made significant progress in medical image analysis, there is still a need for more advanced approaches that can integrate multiple data sources, capture global features, and provide reliable and interpretable results.

2.1.1 System Description

Existing diagnostic systems for respiratory diseases typically follow a structured workflow that includes data acquisition, preprocessing, feature extraction, and classification. In the initial stage, medical images are collected from various sources such as hospitals or publicly available datasets. These images may include chest X-rays or CT scans, depending on the system design.

Once the data is collected, preprocessing techniques are applied to improve image quality and ensure consistency. Common preprocessing steps include resizing images to a standard resolution, normalizing pixel values, and applying data augmentation techniques such as rotation, flipping, and scaling. These steps help in enhancing the robustness of the model and reducing overfitting.

In traditional machine learning systems, feature extraction is performed manually using techniques such as edge detection, texture analysis, and statistical methods. These features are then fed into classification algorithms to identify the presence of disease. However, this approach is limited by the quality of manually extracted features and may not capture all relevant information.

Deep learning-based systems, particularly those using CNNs, automate the feature extraction process. CNNs consist of multiple layers, including convolutional layers, pooling layers, and fully connected layers. These layers work together to learn hierarchical representations of the input image. Lower layers capture basic features such as edges and textures, while deeper layers capture more complex patterns related to disease characteristics.

After feature extraction, the classification stage assigns labels to the input images based on learned patterns. The output typically represents the probability of different disease classes. In respiratory disease detection, common classes include normal, pneumonia, tuberculosis, and COVID-19.

While this pipeline has proven effective, it has certain limitations. The reliance on a single modality restricts the system's ability to capture diverse information.

Additionally, CNN-based models may require large amounts of labeled data for training, which may not always be available in medical domains.

To address these challenges, recent research has focused on developing more advanced architectures that can handle multimodal data and capture global dependencies within images. This has led to the emergence of transformer-based models, which offer new possibilities for improving diagnostic performance.

2.2 Research Review

The field of medical image analysis has witnessed significant advancements with the introduction of deep learning techniques. Numerous research studies have explored the application of these techniques for respiratory disease detection, leading to improved diagnostic accuracy and efficiency.

One of the early approaches involved the use of CNN-based architectures for detecting pneumonia from chest X-ray images. These models demonstrated high accuracy by learning spatial features directly from the images. However, they were limited by their inability to capture long-range dependencies and global context within the image.

With the growing interest in transformer-based models, researchers began exploring the use of Vision Transformers (ViTs) for image classification tasks. Vision Transformers treat images as sequences of patches and use self-attention mechanisms to model relationships between different parts of the image. This approach allows the model to capture global features more effectively than traditional CNNs.

Several studies have shown that Vision Transformers can outperform CNN-based models in medical image classification tasks, particularly when large datasets are available. These models are capable of learning complex patterns and relationships that are crucial for accurate disease detection.

In addition to single-modality approaches, multimodal learning has gained attention as a promising direction for improving diagnostic performance. Researchers have explored combining different types of medical data, such as CT scans, X-rays, and clinical reports, to create more comprehensive models. By integrating multiple sources of information, these models can capture complementary features that may not be available in a single modality.

For example, some studies have proposed dual-branch architectures where separate models are used to process different modalities, and their outputs are combined for final classification. This approach has been shown to improve accuracy and robustness compared to single-modality models.

Another important area of research is feature fusion, which involves combining information from different modalities in a meaningful way. Techniques such as concatenation, attention-based fusion, and weighted averaging have been used to integrate features from multiple sources. Among these, attention-based fusion has shown promising results due to its ability to focus on the most relevant features.

Researchers have also explored the use of hybrid models that combine CNNs and transformers to leverage the strengths of both architectures. CNNs are effective at capturing local features, while transformers excel at modeling global relationships. By combining these approaches, hybrid models can achieve better performance in complex tasks such as medical image analysis.

In recent studies, multimodal Vision Transformer frameworks have been proposed for respiratory disease classification. These models process CT scans and chest X-ray images separately and then combine their features for final prediction. Experimental results from such studies indicate that multimodal approaches significantly outperform traditional methods in terms of accuracy, precision, recall, and F1-score.

Furthermore, advancements in training techniques, such as transfer learning and data augmentation, have helped improve the performance of deep learning models in medical imaging. Transfer learning allows models to leverage knowledge from pre-trained networks, reducing the need for large labeled datasets. Data augmentation techniques help increase dataset diversity and improve generalization.

Overall, the research community has made substantial progress in developing advanced models for respiratory disease detection. However, there is still room for improvement in terms of model efficiency, interpretability, and real-world applicability.

2.3 Gap Analysis

Despite significant advancements in medical image analysis, several gaps remain in the existing literature and systems for respiratory disease diagnosis. Addressing these gaps is essential for developing more effective and reliable diagnostic solutions.

One of the primary gaps is the limited use of multimodal data in existing systems. While many studies focus on either chest X-rays or CT scans, few approaches effectively combine both modalities. Since each imaging technique provides unique information, relying on a single modality can result in incomplete analysis and reduced accuracy.

Another gap lies in the limitations of traditional CNN-based models. Although CNNs have been successful in many applications, they primarily focus on local feature extraction and may not capture global relationships within complex medical images. This can lead to suboptimal performance in tasks that require understanding of spatial dependencies.

Data scarcity and variability also present significant challenges. Medical datasets are often limited in size and may vary in quality and distribution. This can affect the generalization ability of models and lead to performance degradation when applied to new data.

Additionally, many existing models lack interpretability, making it difficult for healthcare professionals to trust their predictions. Providing explanations for model decisions is crucial in medical applications, where accuracy and reliability are of utmost importance.

Another gap is the lack of real-time and scalable solutions. Many research models are developed and tested in controlled environments but may not be suitable for deployment in real-world clinical settings. Factors such as computational requirements, processing time, and integration with existing healthcare systems need to be considered.The proposed project addresses these gaps by introducing a Multimodal Vision Transformer-based approach. By integrating CT scans and chest X-ray images, the system leverages complementary information from multiple sources. The use of transformer-based architecture enables better modeling of global relationships within images, improving diagnostic accuracy.

Furthermore, the proposed system incorporates advanced preprocessing techniques, efficient training strategies, and robust evaluation methods to enhance performance and reliability. By addressing the limitations of existing systems, this project aims to provide a more comprehensive and effective solution for respiratory disease diagnosis.

CHAPTER 3
SYSTEM ANALYSIS & DESIGN

3.1 System Architecture

3.1.1 Architecture Diagram

The The architecture of the proposed Multimodal Vision Transformer (MViT) system is designed to effectively analyze and integrate information obtained from two different medical imaging modalities: chest X-ray images and CT scan slices. The complete framework follows a dual-branch architecture in which each modality is processed independently through a dedicated Vision Transformer pipeline before their learned representations are combined for final disease classification. This multimodal design enables the system to capture complementary diagnostic information from both imaging sources, thereby improving overall prediction performance and robustness.

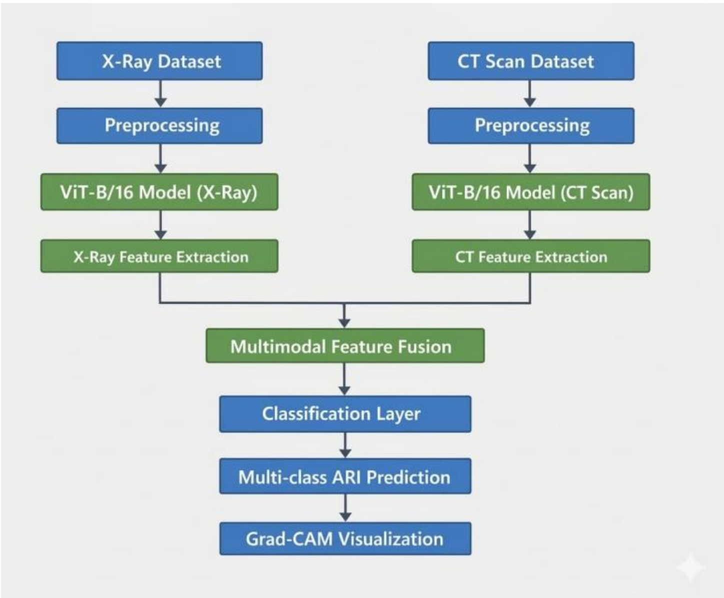

Figure 3.1: System Architecture Diagram - Dual-Branch Multimodal Vision Transformer

The architecture consists of two parallel processing branches. The left branch is responsible for processing chest X-ray images, while the right branch processes CT scan images independently. Although both branches are based on the same Vision Transformer architecture, they do not share parameters or weights. Separate weight matrices are maintained for each branch so that the model can learn modality-specific features effectively. This separation is important because chest X-rays and CT scans differ significantly in terms of image texture, anatomical visibility, resolution, and feature distribution. By training each branch independently, the model becomes capable of extracting specialized and meaningful representations from each imaging modality.

In the first stage of the architecture, input images from both modalities undergo preprocessing operations such as resizing, normalization, noise reduction, and image enhancement. These preprocessing steps ensure uniform image quality and dimensional consistency before feeding the data into the transformer network. After preprocessing, each image is divided into multiple fixed-size patches. These patches are then flattened and transformed into vector embeddings through a linear projection layer. Positional embeddings are added to preserve spatial information that would otherwise be lost during the patching process.

The embedded image patches are subsequently passed through multiple transformer encoder layers. Each encoder layer consists of multi-head self-attention mechanisms and feed-forward neural networks. The self-attention mechanism enables the model to focus on important regions within the medical images by learning long-range dependencies and contextual relationships between patches. This capability is particularly useful in medical imaging applications where abnormalities may appear in subtle or spatially distant regions of the image.

As the data progresses through the transformer layers, high-level semantic representations are generated separately for X-ray and CT modalities. These extracted feature vectors are then forwarded to a feature fusion module. The fusion layer combines the learned representations from both branches into a unified multimodal feature space. This fusion process allows the system to integrate complementary information obtained from X-rays and CT scans, improving diagnostic accuracy compared to single-modality approaches.

Following feature fusion, the combined feature representation is passed through fully connected dense layers and a final classification head. The classification layer predicts the target disease category, such as normal, pneumonia, tuberculosis, COVID-19, or other thoracic conditions depending on the dataset used for training. Softmax activation is employed in the output layer to generate probability scores for each class.

The proposed architecture offers several advantages. First, the independent dual-branch structure prevents interference between modalities and allows specialized feature extraction. Second, the Vision Transformer backbone improves global feature learning through attention mechanisms. Third, multimodal fusion enhances the reliability and generalization capability of the model by leveraging complementary diagnostic information from multiple imaging sources. Overall, the architecture provides an efficient and scalable framework for advanced medical image classification tasks.

3.1.2 Workflow Description

The Processing Pipeline of the Proposed System

The proposed Multimodal Vision Transformer system follows a structured and sequential processing pipeline designed to efficiently extract, integrate, and classify diagnostic information from chest X-ray and CT scan images. The pipeline consists of eight major stages, beginning with data ingestion and ending with interpretable visualization of model predictions. Each stage contributes to improving feature representation, classification accuracy, and explainability of the overall system.

Stage 1 – Data Ingestion

In the first stage, the system receives paired medical imaging data corresponding to a single patient case. Each pair consists of two complementary imaging modalities:

- A chest X-ray image
- A representative two-dimensional (2D) slice extracted from the patient's CT scan volume

The use of paired modalities enables the model to learn complementary diagnostic information that may not be visible in a single imaging source alone. Chest X-rays provide quick and broad structural information, while CT scans offer more detailed cross-sectional anatomical features.

Once the paired images are obtained, they are independently routed into their dedicated processing branches within the dual-branch architecture. The X-ray image is sent to the X-ray Vision Transformer branch, whereas the CT scan slice is sent to the CT Vision Transformer branch. This separation allows each branch to specialize in learning modality-specific patterns and characteristics.

Stage 2 – Preprocessing

Before feature extraction, both imaging modalities undergo a preprocessing phase to ensure consistency and compatibility with the transformer architecture. Since medical images may originate from different hospitals, devices, and datasets, preprocessing is essential for standardizing the inputs.

The preprocessing operations include:

- Resizing all images to a fixed resolution of **224 × 224 pixels**
- Normalizing pixel intensity values to the range **[0, 1]**

Normalization is achieved by dividing pixel values by 255:

$$x_{normalized} = \frac{x}{255}$$

This normalization process stabilizes gradient updates during training and improves convergence of the optimization algorithm. Standardized image dimensions also ensure compatibility with fixed patch generation in the Vision Transformer architecture.

Additional preprocessing steps such as noise reduction, contrast enhancement, or data augmentation may also be incorporated to improve image quality and increase model generalization capability.

Stage 3 – Patch Embedding

After preprocessing, each image is divided into smaller non-overlapping patches. In this system, the image size is **224 × 224**, and the patch size is **16 × 16** pixels.

The total number of patches generated per image is calculated as:

$$\left(\frac{224}{16}\right)^2 = 14^2 = 196$$

Thus, each image produces **196 patches**.

Each patch is flattened into a one-dimensional vector. For RGB images, the flattened vector dimension becomes:

$$16 \times 16 \times 3 = 768$$

These flattened vectors are then projected into the transformer embedding space using a learnable linear projection layer. Since transformers do not inherently preserve spatial ordering, positional embeddings are added to every patch token. Positional embeddings encode the spatial location of patches within the image, enabling the model to understand spatial relationships between image regions.

This stage converts raw pixel information into structured sequential embeddings suitable for transformer-based processing.

Stage 4 – Transformer Encoding

The embedded patch tokens are passed through multiple stacked Vision Transformer encoder blocks. Each encoder block contains the following components:

- Layer Normalization
- Multi-Head Self-Attention (MHA)
- Feed-Forward Network (FFN)
- Residual Skip Connections

The most important component is the multi-head self-attention mechanism, which enables the model to learn contextual relationships between all image patches simultaneously. Unlike convolutional neural networks that focus primarily on local neighborhoods, self-attention captures long-range spatial dependencies across the entire image.

The attention operation can be mathematically represented as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where:

- Qrepresents query vectors
- Krepresents key vectors
- Vrepresents value vectors
- $d_k$is the dimensionality scaling factor
    - This mechanism allows the model to focus on diagnostically important regions such as lung opacities, lesions, infections, or abnormal tissue structures.

Stage 5 – Feature Extraction

Following transformer encoding, the network produces high-level semantic feature representations for each modality. A special learnable token called the **class token ([CLS])** is prepended to the sequence before encoding begins.

After passing through all transformer layers, the final state of this class token acts as a compact global representation of the entire image. It summarizes the important semantic information extracted from all patches.

Separate class token vectors are generated for:

- The chest X-ray branch
- The CT scan branch

These vectors contain modality-specific diagnostic information learned independently by each transformer encoder.

Stage 6 – Multimodal Fusion

Once features are extracted from both branches, the model performs multimodal fusion to combine information from the two imaging modalities.

The fusion process is implemented using feature concatenation:

$$F_{fusion} = [F_{Xray}; F_{CT}]$$

where:

- F_{Xray}= feature vector from the X-ray branch
- F_{CT}= feature vector from the CT branch

Concatenation preserves the full feature information from both modalities without compression or averaging. This strategy enables the classifier to simultaneously analyze complementary imaging characteristics obtained from X-rays and CT scans.

The fused representation forms a comprehensive multimodal feature space capable of improving disease discrimination performance.

Stage 7 – Classification

The fused feature vector is passed to the final classification head. This stage consists of:

- One or more fully connected dense layers
- Dropout regularization (optional)
- Softmax activation function

The softmax layer converts raw output scores into probability values corresponding to the target disease classes.

The softmax function is defined as:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^{n} e^{z_j}}$$

where:

- z_i is the output score for class i
- n is the total number of classes

The disease category with the highest predicted probability is selected as the final model prediction. The system can classify diseases such as:

- Normal
- Pneumonia
- Tuberculosis
- COVID-19

or other thoracic abnormalities depending on the dataset used.

Stage 8 – Visualization and Explainability

The final stage focuses on model interpretability and explainability, which are essential in medical AI systems. During inference, explainable AI techniques are used to highlight image regions that most strongly influenced the model's prediction.

The proposed system incorporates:

- Grad-CAM (Gradient-weighted Class Activation Mapping)
- Occlusion Sensitivity Analysis

Grad-CAM generates heatmaps by analyzing gradient information flowing through the network. These heatmaps identify important pathological regions such as infected lung areas or lesions.

Occlusion sensitivity works by systematically masking parts of the image and observing changes in prediction confidence. Regions causing significant prediction changes are considered highly important.

These visualization techniques improve clinician trust, assist in diagnosis verification, and enhance transparency of the deep learning model by providing interpretable visual explanations for predictions.

3.1.3 Data Flow

The The data flow within the proposed Multimodal Vision Transformer system follows a systematic sequence of operations that transforms raw medical images into final disease predictions with interpretable visual explanations. The complete workflow ensures efficient feature extraction, multimodal learning, classification, and optimization during both training and inference phases.

Initially, raw chest X-ray images and CT scan slices are collected from the medical dataset repository. Each patient sample contains paired imaging data representing the two modalities. These images are first routed through the preprocessing pipeline, where several standardization operations are performed. The preprocessing stage includes image resizing, normalization, and formatting to ensure consistency across different datasets and imaging devices. All images are resized to a uniform resolution of **224 × 224 pixels**, and pixel intensity values are normalized to the range **[0, 1]** to improve numerical stability during model training.

After preprocessing, the images are independently forwarded into their corresponding Vision Transformer branches. The chest X-ray image enters the X-ray transformer branch, while the CT scan slice enters the CT transformer branch. Inside each branch, the image is divided into fixed-size patches, converted into patch embeddings, and combined with positional embeddings. These embedded patch sequences are then processed through multiple transformer encoder layers consisting of multi-head self-attention mechanisms and feed-forward neural networks

During transformer encoding, the model learns complex spatial relationships and high-level semantic patterns present in the medical images. The encoded outputs generated by both branches are represented through class token feature vectors that summarize the global content of each modality. These feature representations are subsequently forwarded to the multimodal fusion layer.

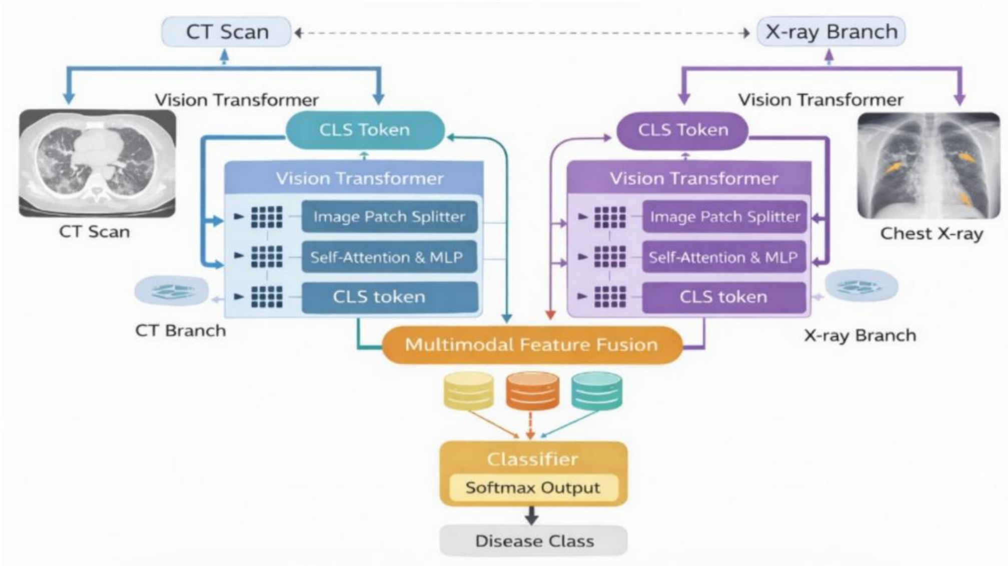

Figure 3.2: Data Flow Diagram - From Input Images to Multi-class Disease Prediction

In the fusion stage, the feature vectors obtained from the CT branch and X-ray branch are concatenated to form a unified multimodal representation. This fused feature vector combines complementary diagnostic information from both imaging modalities, enabling the model to make more accurate and reliable predictions compared to single-modality systems.

The fused representation is then passed into the classification head, which consists of fully connected dense layers followed by a softmax activation function. The classifier produces predicted disease labels along with associated probability scores for each disease category. The class with the highest probability value is selected as the final prediction result.

Along with the prediction output, the system also generates explainability visualizations using Grad-CAM and occlusion sensitivity techniques. These visualization methods create heatmaps that highlight the most influential regions of the CT scan and chest X-ray images contributing to the model's decision. Such visual explanations improve interpretability and help clinicians understand the reasoning behind predictions.

During the training phase, the predicted probability distributions are compared with the corresponding ground truth labels using the categorical cross-entropy loss function. The loss function measures the difference between predicted outputs and actual class labels, guiding the learning process of the model.

The categorical cross-entropy loss is mathematically represented as:

$$L = -\sum_{i=1}^{n} y_i \log(\hat{y}_i)$$

where:

- $y_i$represents the true class label
- $\hat{y}_i$represents the predicted probability for class i
- $n$represents the total number of classes

The optimization of model parameters is performed using the Adam optimizer, which adaptively updates network weights through backpropagation. Adam combines the advantages of momentum and adaptive learning rate optimization, enabling faster convergence and stable training performance.

Through iterative forward propagation, loss computation, and backpropagation, the model gradually learns optimal feature representations for accurate multimodal disease classification. This complete data flow pipeline ensures efficient learning, robust prediction performance, and enhanced interpretability for medical image analysis applications.

3.2 System Design

3.2.1 Design Approach

The design philosophy underlying the proposed system prioritizes modularity, interpretability, and deployment feasibility. The dual-branch architecture was chosen over a single-branch approach to ensure that each modality is processed by a dedicated network with the capacity to learn modality-specific features without interference from the other input type. This separation reflects the distinct statistical properties of CT and X-ray images, which differ fundamentally in their contrast mechanisms, noise characteristics, and clinically relevant feature distributions.

The ViT-B/16 variant was selected as the backbone for each branch. This variant uses a patch size of 16x16 pixels and a base-scale model with 12 transformer encoder layers, 12 attention heads, and a hidden dimension of 768. It provides an optimal balance between representational capacity and computational efficiency for this application domain.

The use of independent branch weights rather than shared weights was a deliberate design decision. Sharing weights between branches would implicitly assume that CT and X-ray images share the same feature space, which is not the case. Independent weights allow each branch to develop a specialized representation adapted to the statistical properties of its input modality.

3.2.2 Database Design

The system does not use a traditional relational database for data storage. Instead, the dataset is organized on the file system in a structured directory hierarchy. The base dataset directory contains subdirectories for each disease category (COVID-19, Pneumonia, Tuberculosis, Normal). Within each category directory, separate subdirectories store CT scan images and chest X-ray images. A metadata CSV file records the pairing of CT and X-ray image filenames for each patient case, the associated disease label, and the data split assignment (train/validation/test).

3.2.3 UML Diagrams

The Use Case Diagram of the proposed Respiratory Disease Classification System illustrates the interaction between the end user and the AI-powered diagnostic framework. The system is primarily designed to assist radiologists in analyzing multimodal medical images, specifically chest X-rays and CT scan slices, for automated respiratory disease detection and classification.

The diagram involves two primary actors:

- Radiologist (End User)
- AI Model (Multimodal Vision Transformer System)

The radiologist acts as the primary user of the system. The workflow begins when the radiologist uploads a paired set of medical images consisting of:

- A chest X-ray image
- A CT scan slice

These images are submitted through the inference interface of the system. Once the images are uploaded, the AI Model automatically initiates the processing pipeline.

The first operation performed by the AI system is preprocessing, where the uploaded images undergo resizing, normalization, and formatting. The preprocessed images are then forwarded to their respective Vision Transformer branches for feature extraction.

Within the AI Model, the following major tasks are executed:

1. **Image Preprocessing**
   Standardizes the uploaded CT and X-ray images for consistent analysis.
2. **Feature Extraction**
   The Vision Transformer branches independently learn modality-specific features from CT and X-ray inputs.
3. **Multimodal Fusion**
   The extracted features from both modalities are combined into a unified feature representation.
4. **Disease Classification**
   The classifier predicts the respiratory disease category and generates probability scores.

5. **Explainability Visualization**
   Grad-CAM heatmaps are generated to highlight diagnostically important regions influencing the prediction.

After processing is completed, the AI system outputs:

- Predicted disease label
- Prediction probability scores
- Grad-CAM visualization heatmaps

The radiologist reviews these outputs along with the highlighted image regions to support clinical interpretation and decision-making. Although the system provides automated predictions, the final clinical diagnosis remains under the control of the radiologist, ensuring human oversight and accountability.

The Use Case Diagram therefore demonstrates a collaborative interaction between artificial intelligence and medical professionals, where AI functions as an intelligent decision-support tool rather than a replacement for clinical expertise.

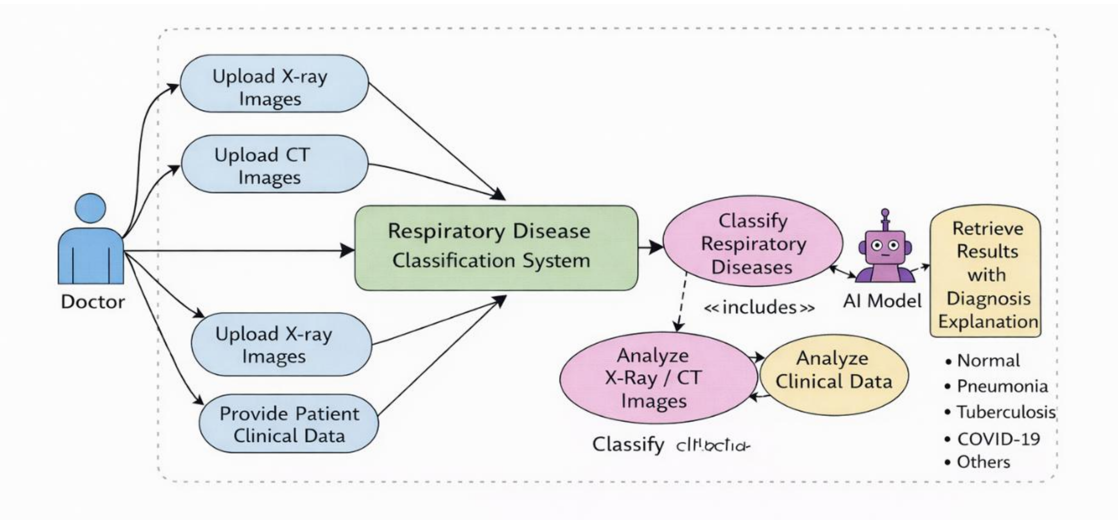

Figure 3.3: Use Case Diagram — Respiratory Disease Classification System

The Sequence Diagram illustrates the temporal flow of interactions between the system components during an inference request. It represents how data and control signals move through different modules of the proposed Multimodal Vision Transformer framework from image submission to prediction generation.

The inference process begins when the radiologist submits paired CT and chest X-ray images through the system interface. This request triggers a sequence of operations across multiple interconnected modules.

The interaction sequence proceeds as follows:

Step 1 – Image Submission

The radiologist uploads the paired medical images into the system using the graphical inference interface. The uploaded images are forwarded to the preprocessing module for standardization.

Step 2 – Preprocessing Module

The preprocessing module performs several operations on the input images, including:

Image resizing to 224 × 224 pixels

Pixel normalization

Noise handling and formatting

After preprocessing, the images are transformed into a suitable format for Vision Transformer processing.

Step 3 – Parallel Processing in ViT Branches

The preprocessed CT scan image and chest X-ray image are independently routed to their respective Vision Transformer branches:

- CT ViT Branch
- X-ray ViT Branch

Both branches execute in parallel to improve computational efficiency and preserve modality-specific feature learning. Each branch performs:

- Patch extraction
- Patch embedding
- Positional encoding

- Transformer-based feature extraction using self-attention mechanisms

The transformer encoders generate high-level semantic feature representations for each modality.

Step 4 – Feature Fusion

The extracted feature vectors from the CT and X-ray branches are transmitted to the fusion module. The fusion layer concatenates both modality representations into a single multimodal feature vector.

This fused representation combines complementary diagnostic information from both imaging modalities, improving disease classification capability.

Step 5 – Classification

The fused feature vector is passed into the classification module. The classifier applies dense layers and softmax activation to compute probability scores for each disease category.

The output generated by the classifier includes:

- Predicted disease class
- Probability distribution across all classes

The disease with the highest probability is selected as the final prediction.

Step 6 – Grad-CAM Visualization

Simultaneously, the visualization module computes Grad-CAM heatmaps based on the model's prediction. The Grad-CAM algorithm identifies the image regions that contributed most strongly to the classification decision.

These heatmaps provide visual interpretability and assist clinicians in understanding the reasoning process of the AI system.

Step 7 – Result Display

Finally, the system displays the following outputs to the radiologist:

- Disease prediction
- Confidence probability scores

- Grad-CAM visualization maps

The radiologist reviews the prediction results and interpretability outputs before making the final clinical assessment.

The Sequence Diagram effectively demonstrates the coordinated interaction between preprocessing, transformer encoding, multimodal fusion, classification, and explainability modules during real-time inference.

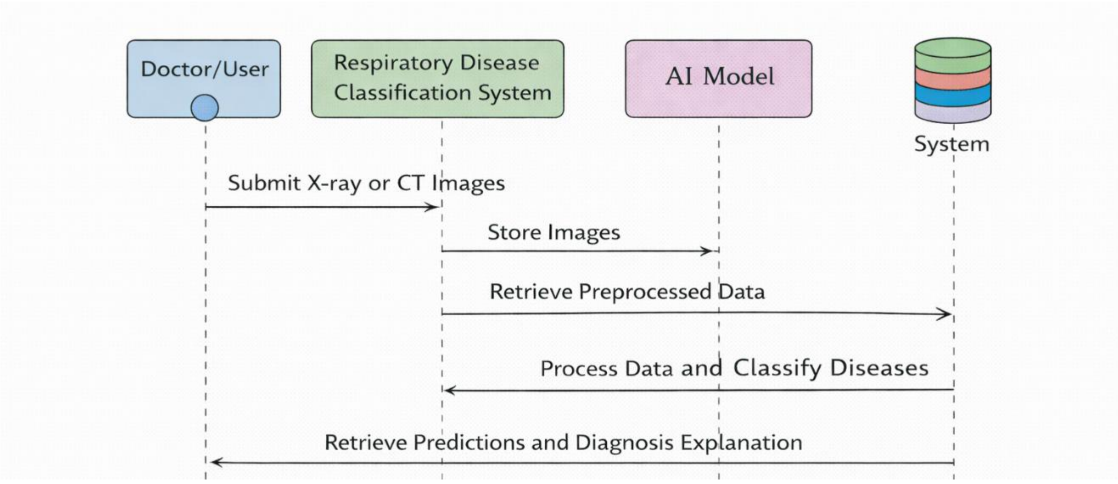

Figure 3.4: Sequence Diagram — Image Submission to Prediction Output

3.3 Module Description

Table 3.1: System Module Descriptions and Responsibilities

Module	Name	Responsibility
Module 1	Data Preprocessing	Image loading, resizing to 224x224, normalization, CT slice extraction, dataset balancing, train/val/test split
Module 2	ViT Branch — X-ray	Patch extraction, positional embedding, transformer encoding, class token extraction for X-ray inputs

Module 3	ViT Branch — CT Scan	Patch extraction, positional embedding, transformer encoding, class token extraction for CT inputs
Module 4	Multimodal Feature Fusion	Concatenation of X-ray and CT class token vectors, generation of unified fused representation
Module 5	Classification Head	Dense layer + Softmax for multi-class prediction across COVID-19, Pneumonia, Tuberculosis, Normal
Module 6	Interpretability Module	Grad-CAM heatmap generation and occlusion sensitivity analysis for visual explanation
Module 7	Training Pipeline	Adam optimization, categorical cross-entropy loss, early stopping, model checkpointing
Module 8	Evaluation Module	Computation of Accuracy, Precision, Recall, F1 Score; confusion matrix generation

CHAPTER 4
IMPLEMENTATION

4.1 Technologies Used

4.1.1 Software Tools

The entire implementation was carried out using Python 3.10 as the primary programming language. Python was selected due to its extensive ecosystem of scientific computing, deep learning, and medical image processing libraries, as well as its broad adoption in the research community, which ensures reproducibility and maintainability of the codebase.

The deep learning model was developed using TensorFlow 2.12 with the Keras high-level API. TensorFlow provides efficient GPU-accelerated computation for training deep neural networks, while Keras simplifies model construction through a modular layer-based interface. The custom ViT layers were implemented using TensorFlow's functional API to enable flexible model composition.

The Kaggle platform was used as the primary training environment due to its provision of free GPU compute resources (NVIDIA Tesla P100) and its direct integration with the source datasets. Experiments were also conducted on Google Colab as a secondary environment.

4.1.2 Software and Hardware Requirements

Table 4.1: Software and Hardware Configuration

Component	Specification
GPU	NVIDIA Tesla P100 (16 GB VRAM) — Kaggle Cloud
CPU	Intel Xeon (2 cores) — Kaggle Cloud
RAM	16 GB — Kaggle Cloud
Storage	SSD — 30 GB project workspace
OS	Linux Ubuntu 20.04
Python	3.10

TensorFlow	2.12.0
Keras	Built-in (tf.keras)

4.1.3 Frameworks and Libraries

The following libraries were used in the implementation:

- TensorFlow / Keras — Model construction, training, and evaluation
- NumPy — Numerical operations and array manipulation
- OpenCV (cv2) — Image loading, resizing, and preprocessing
- scikit-learn — Metrics computation and data splitting utilities
- nibabel — NIfTI format loading for 3D CT scan volumes
- pandas — Dataset metadata management and CSV operations
- matplotlib — Visualization of training curves, confusion matrices, and heatmaps
- tqdm — Progress monitoring during dataset loading and training

4.2 Algorithms and Models

4.2.1 Dataset Preparation

Table 4.2: Dataset Distribution Across Disease Categories

Disease Category	X-ray Images	CT Images
COVID-19	2,500	1,500
Pneumonia	3,000	1,200
Tuberculosis	2,000	800
Normal (Other)	2,500	1,200
Total	10,000	4,700

Images were sourced from six publicly available Kaggle datasets including the COVID-19 Radiography Dataset, the Tuberculosis Chest X-ray Dataset, the RSNA Pneumonia Detection Dataset, and the SARS-CoV-2 CT Scan Dataset. For 3D CT scan volumes, 2D axial slices demonstrating the most prominent pulmonary involvement were selected. All images were resized to 224x224 pixels and normalized by dividing pixel intensity values by 255. Data augmentation techniques including random horizontal flipping, rotation within ±15 degrees, and brightness jitter were applied to the training set to improve model generalization. The dataset was split into 80% training, 10% validation, and 10% testing.

4.2.2 Algorithm Description — Vision Transformer

The Vision Transformer operates on the principle of treating an image as a sequence of patch tokens, analogous to how natural language transformers process word tokens. For an input image I of dimensions H x W x C, where H and W are the height and width and C is the number of channels, the image is divided into N non-overlapping patches of size P x P, where:

$$N = (H \times W) / P^2$$

For our configuration with H = W = 224 and P = 16, N = 196 patches. Each patch is flattened into a vector of dimension $P^2C = 768$ and projected into the embedding space through a learned linear projection matrix E:

$$zp = xp \cdot E$$

A learnable class token [CLS] is prepended to the patch sequence, and learnable positional embeddings are added to encode spatial information. The resulting sequence of 197 vectors is processed by L stacked transformer encoder blocks. Each block applies Multi-Head Self-Attention (MHA) followed by a Feed-Forward Network (FFN) with layer normalization and residual connections:

$$\text{Attention}(Q, K, V) = \text{softmax}(QK\hat{}T / \sqrt{dk}) \cdot V$$

where Q, K, and V are the query, key, and value matrices computed from the input feature matrix X through learned weight matrices WQ, WK, and WV respectively. The dk parameter is the key dimension, and the square root scaling prevents vanishing gradients in the softmax operation. Multi-head attention concatenates the outputs of h attention heads and projects the result through an output matrix WO.

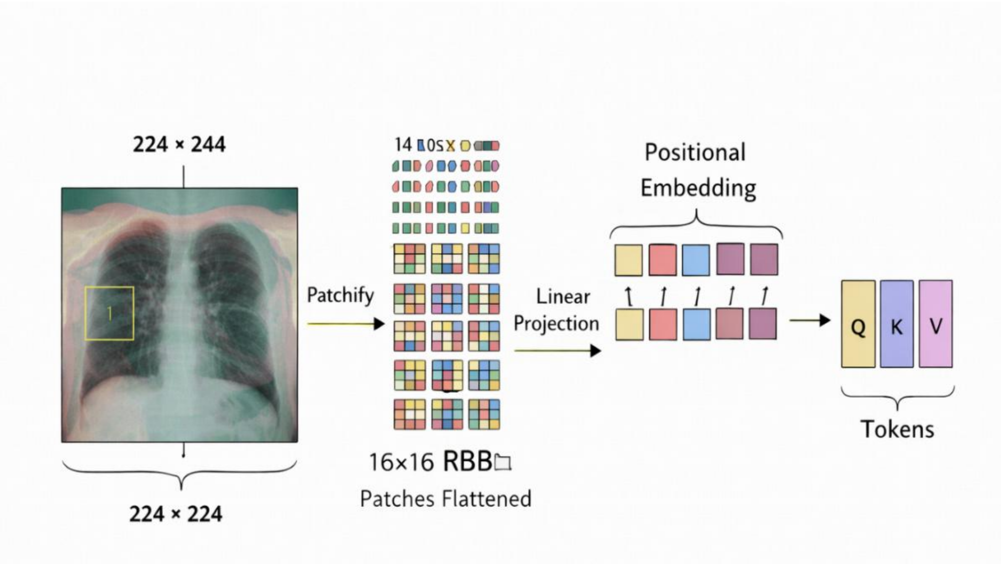

Figure 4.1: Patch Embedding Process - Image to Token Sequence

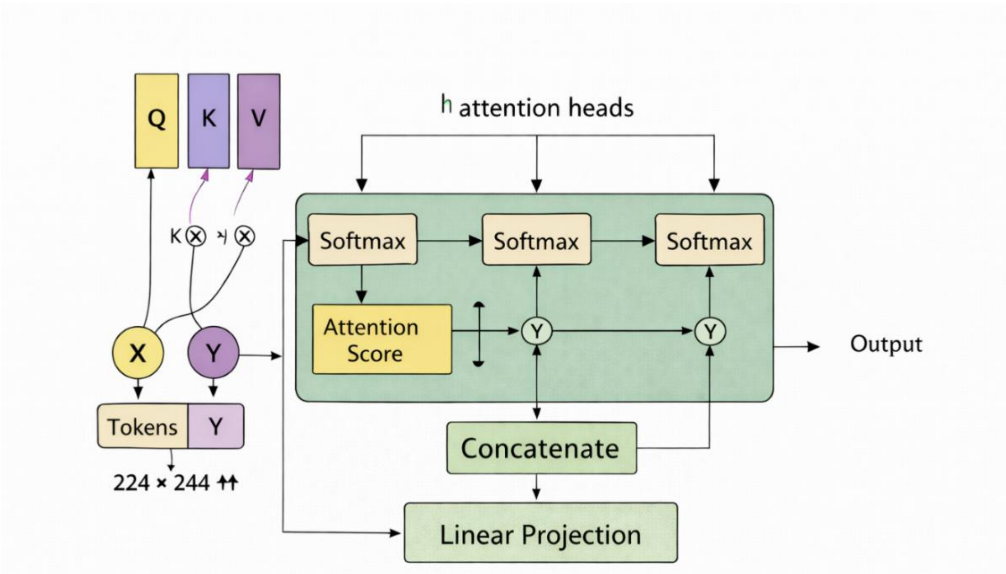

Figure 4.2: Multi-Head Self-Attention Mechanism Diagram

4.2.3 Multimodal Feature Fusion and Classification

After transformer encoding, the **[CLS] token** from each modality (CT and X-ray) represents the global contextual information of the respective image. These tokens capture high-level semantic features learned through the self-attention mechanism, enabling the model to understand complex spatial relationships within medical images.

The fusion process combines these modality-specific representations to leverage complementary information. CT scans provide detailed structural insights, while X-rays offer quick diagnostic patterns. By concatenating both feature vectors, the model gains a more comprehensive understanding of the disease.

$$F_{fusion} = [F_{CT}; F_{XR}]$$

To further enhance feature learning, the fused vector is passed through one or more **fully connected (dense) layers**. These layers help in learning non-linear relationships between combined features and improve discriminative power. Regularization techniques such as **dropout** may also be applied at this stage to reduce overfitting.

The final classification layer uses the **Softmax activation function**, which converts the output logits into probability scores for each class:

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^{C} e^{z_j}}$$

Here, each probability represents the likelihood of the input belonging to a particular disease class (COVID-19, pneumonia, tuberculosis, or normal). The predicted class is determined by selecting the maximum probability value.

To train the model effectively, the **categorical cross-entropy loss function** is used:

$$L = -\sum_{i=1}^{C} y_i \log(\hat{y}_i)$$

This loss function measures the difference between the true labels and predicted probabilities, guiding the model to minimize classification errors during training.

Additionally, the model is optimized using the **Adam optimizer**, which adapts the learning rate dynamically for faster convergence. During training, **batch normalization** and **early stopping** strategies can be applied to stabilize learning and prevent overfitting.

Overall, multimodal feature fusion significantly improves classification performance by:

- Utilizing complementary information from CT and X-ray images
- Enhancing feature representation through combined learning
- Increasing robustness and generalization of the model
- Providing more accurate and reliable disease predictions

Table 4.3: Model Training Parameters

Parameter	**Value**
Optimizer	Adam
Learning Rate	0.0001
Batch Size	16 – 32
Loss Function	Categorical Cross-Entropy
Epochs	Up to 30 (with Early Stopping)
Early Stopping Patience	5 epochs (monitor: val_accuracy)
Model Checkpoint	Save best model based on val_accuracy
Input Image Size	224 x 224 pixels
Patch Size	16 x 16 pixels
Number of Patches	196
Transformer Layers	12
Attention Heads	12
Embedding Dimension	768

4.2.4 Flowchart

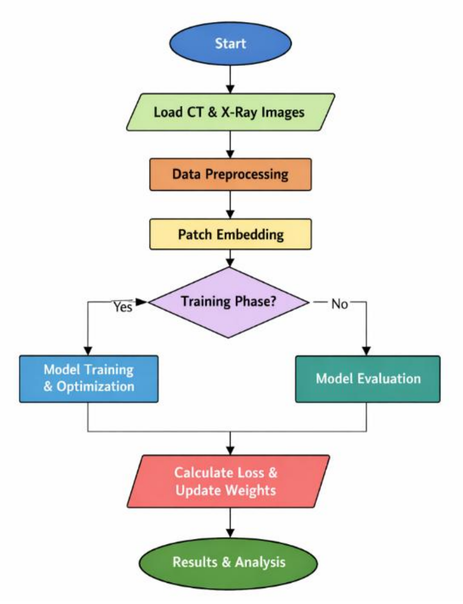

Figure 4.4: Flowchart — Model Training and Evaluation Pipeline

4.3 Working and Screenshots

4.3.1 Input Processing

During the data loading phase, the preprocessing pipeline reads image files from their respective directories, applies resizing and normalization, and assembles them into paired batches. CT scan volumes in NIfTI format are loaded using nibabel, and the most diagnostically relevant axial slice is extracted based on the intensity profile of each volume. The extracted slice is then processed through the same resizing and normalization pipeline as the X-ray images.

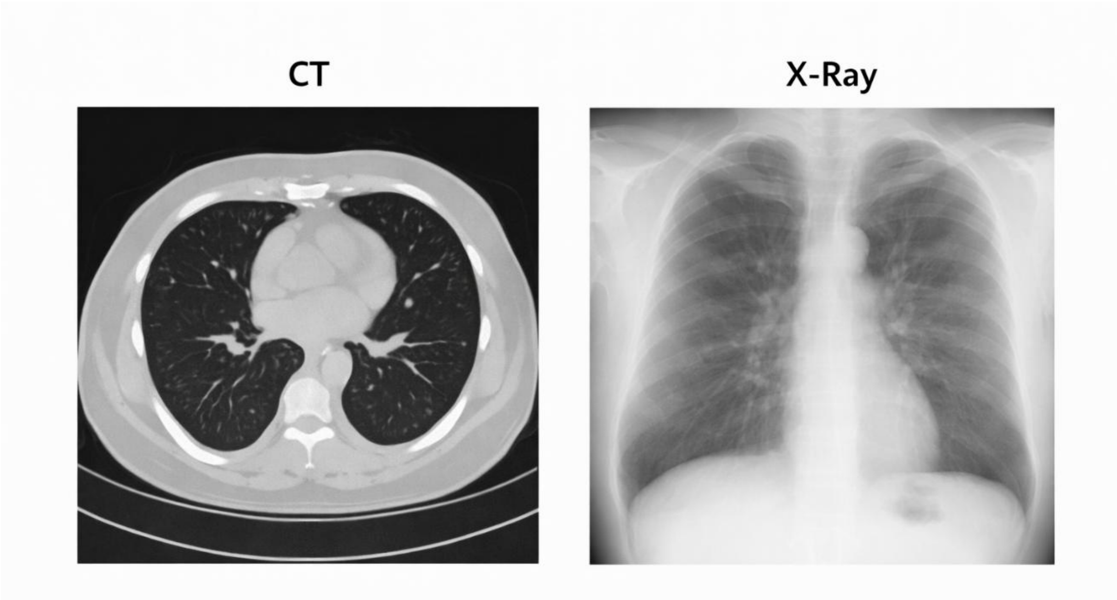

Figure 4.3: Sample Preprocessing Output — Resized and Normalized CT and X-ray Image Pairs

4.3.2 Execution Process

Training is initiated by calling the model.fit() function with the training dataset generator and validation dataset generator as inputs. The training loop runs for each epoch, computing forward passes, loss values, and gradient updates through the Adam optimizer. The ModelCheckpoint callback saves the weights of the best model observed on the validation accuracy metric. The EarlyStopping callback monitors the validation accuracy and terminates training if no improvement is observed for five consecutive epochs, preventing overfitting and unnecessary computation.

4.3.3 Output Generation

Following training, the best model weights are restored and inference is performed on the held-out test dataset. For each test sample, the model outputs a four-element probability vector. The predicted class is the argmax of this vector. The Grad-CAM module computes the gradient of the predicted class score with respect to the feature maps in the final transformer encoder block, producing a coarse localization map. This map is upsampled to the original image dimensions using bilinear interpolation and overlaid on the input image as a color heatmap, highlighting the regions most influential in the model's prediction.

CHAPTER 5
RESULTS AND DISCUSSION

5.1 Outputs

5.1.1 Training and Validation Curves

The training and validation performance curves provide important insights into the learning behavior, convergence characteristics, and generalization capability of the proposed Multimodal Vision Transformer (MViT) framework. During training, both accuracy and loss metrics were continuously monitored across epochs to evaluate model stability and detect

potential overfitting or underfitting issues.

Figure 5.1: Training Accuracy and Validation Accuracy Curves Over Epochs

The training accuracy and validation accuracy curves demonstrated a consistent upward trend throughout the training process. As the number of epochs increased, the model progressively learned meaningful feature representations from the multimodal medical imaging dataset, resulting in improved classification performance. The gradual increase in training accuracy indicates that the Vision Transformer branches successfully captured discriminative spatial and semantic features from both chest X-ray and CT scan modalities.

The validation accuracy closely followed the training accuracy curve throughout the training process. This behavior is particularly significant because it indicates that the model generalized effectively to previously unseen validation samples rather than merely memorizing the training data. The absence of a significant divergence between the two curves confirms that overfitting was successfully minimized.

The convergence behavior of the curves demonstrates stable optimization and efficient learning dynamics. Both curves eventually stabilized at high accuracy values, showing that the model reached an optimal learning state. The final validation accuracy achieved by the proposed framework was approximately **88%**, indicating strong performance in multiclass respiratory disease classification.

The high validation performance confirms that the proposed multimodal fusion strategy effectively combines complementary information from chest X-rays and CT scans, improving the model's diagnostic capability across different disease categories.

The relationship between training and validation accuracy can be interpreted as follows:

- A small gap between the curves indicates good generalization
- Parallel growth trends indicate stable learning
- Absence of sharp fluctuations indicates robust optimization
- Stable convergence suggests effective hyperparameter selection

The early stopping mechanism played a crucial role in improving training efficiency and preventing overfitting. During experimentation, validation accuracy improvements became minimal after approximately **epoch 23**, indicating performance saturation. At this point, early stopping was triggered automatically to terminate further training.

Early stopping prevented the model from continuing unnecessary parameter updates that could potentially lead to memorization of training data and degraded generalization performance. This strategy also reduced computational cost and training time while preserving optimal model weights.

The training and validation loss curves further support the effectiveness of the proposed framework. Both curves exhibited complementary downward trends throughout the training process. Initially, the loss values were relatively high because the model weights were randomly initialized. As training progressed, the loss steadily decreased, indicating continuous improvement in prediction quality.

The training loss decreased as the optimizer minimized classification errors on the training dataset. Simultaneously, the validation loss also declined and eventually stabilized at a low value, demonstrating that the learned representations generalized well to unseen data samples.

The loss minimization process can be represented using categorical cross-entropy loss:

$$L = -\sum_{i=1}^{n} y_i \log(\hat{y}_i)$$

where:

- $y_i$represents the true class label
- $\hat{y}_i$represents the predicted probability
- $n$represents the total number of classes

One of the most important observations from the loss curves is the absence of a pronounced gap between training loss and validation loss. In deep learning systems, a large gap typically indicates overfitting, where the model performs well on training data but poorly on unseen samples. However, in the proposed framework, both curves remained closely aligned, confirming that the model maintained strong generalization capability.

Several factors contributed to this stable learning behavior:

- Data augmentation techniques improved dataset diversity
- Early stopping prevented excessive training
- Multimodal learning improved feature robustness
- Transformer attention mechanisms enhanced global feature extraction
- Proper normalization stabilized optimization

The Adam optimizer further contributed to smooth convergence by adaptively adjusting learning rates during training. The optimizer efficiently minimized loss while maintaining training stability across epochs.

Overall, the training and validation curves demonstrate that the proposed Multimodal Vision Transformer framework achieved efficient convergence, strong generalization performance, and stable optimization behavior. The results confirm that the system effectively learns meaningful and clinically relevant feature representations from multimodal medical images while successfully avoiding overfitting. These findings validate the robustness and reliability of the proposed architecture for automated respiratory disease classification.

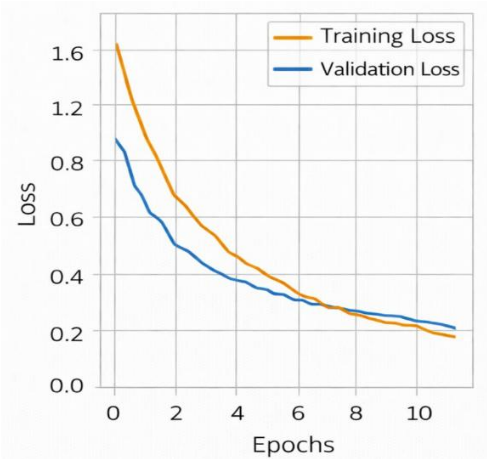

Figure 5.2: Training Loss and Validation Loss Curves Over Epochs

5.1.2 Confusion Matrix

The confusion matrix for the test dataset provides a detailed view of the model's per-class classification behavior. The matrix shows that the model correctly classified the majority of test samples across all four categories, with the most frequent misclassifications occurring between Pneumonia and COVID-19

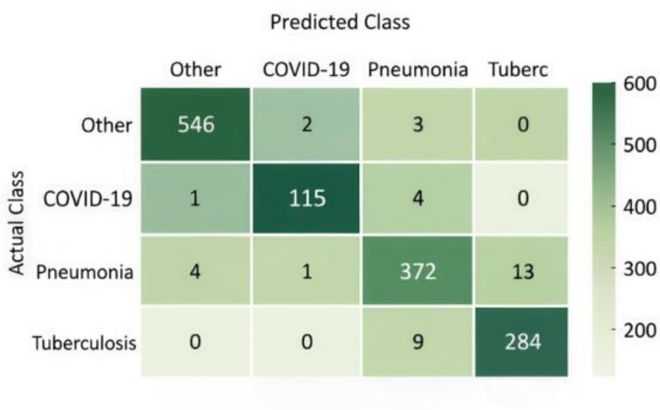

Figure 5.3: Confusion Matrix — Multi-Class Disease Classification on Test Dataset

,which is consistent with the known radiological similarity of these two conditions. Normal cases were classified with the highest accuracy, reflecting the visual distinctiveness of healthy lung images

5.1.3 Grad-CAM Visualization

The Grad-CAM heatmaps demonstrate that the model's attention is appropriately localized to clinically relevant regions. For COVID-19 cases, the heatmaps highlight bilateral peripheral ground-glass opacities. For Tuberculosis cases, attention is concentrated in the upper lung zones where cavitations and consolidations are characteristically located. For Pneumonia, the heatmaps highlight areas of focal consolidation. These patterns are consistent with established radiological diagnostic criteria, providing evidence that the model has learned clinically meaningful representations.

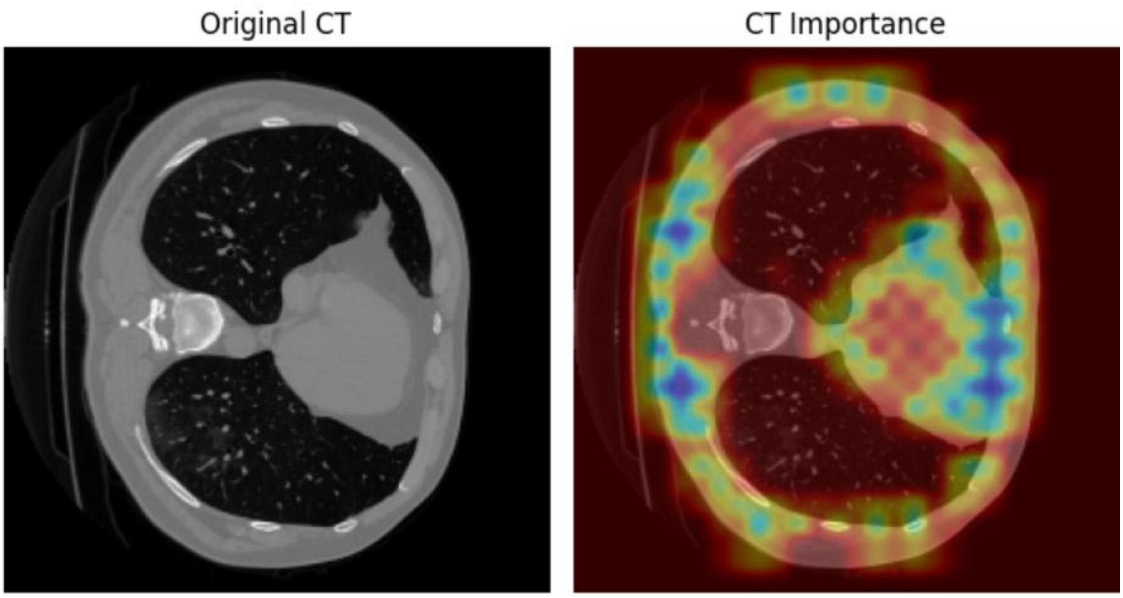

Figure 5.4: Grad-CAM Heatmap Visualization on Sample CT Scan Images for Each Disease Class

5.2 Performance Analysis

5.2.1 Metrics

The model was evaluated using four standard classification metrics computed on the held-out test dataset. Accuracy measures the overall proportion of correctly classified samples. Precision measures the proportion of positive predictions that were correct, reflecting the model's ability to avoid false positives. Recall measures the proportion of actual positive cases that were correctly detected, reflecting the model's sensitivity. The F1 Score is the harmonic mean of Precision and Recall, providing a balanced measure of classification quality.

5.2.2 Evaluation Results

Table 5.1: Per-Class Performance Metrics — Precision, Recall, and F1 Score

Disease Class	Precision (%)	Recall (%)	F1 Score (%)
COVID-19	91	89	90
Pneumonia	86	84	85
Tuberculosis	88	90	89
Normal (Other)	92	90	91
Overall Average	89.25	88.25	88.75

Table 5.2: Overall Model Accuracy Summary

Metric	Value
Overall Validation Accuracy	~88%
Overall Test Accuracy	~87.5%
Macro-Average Precision	89.25%
Macro-Average Recall	88.25%
Macro-Average F1 Score	88.75%

5.2.3 Observations

The experimental results demonstrate that the proposed Multimodal Vision Transformer (MViT) framework achieves strong, stable, and reliable classification performance across all four disease categories. The integration of chest X-ray and CT scan modalities significantly enhances the model's ability to capture complementary diagnostic features, leading to improved overall accuracy and robustness compared to single-modality approaches.

Among all disease categories, the **Normal** class achieved the highest performance with an **F1-score of 91%**. This strong result can be attributed to the relatively clear and visually distinguishable characteristics of healthy lung images. Normal chest scans generally exhibit well-defined lung structures without opacities, consolidations, or abnormal tissue patterns, allowing the transformer model to easily learn discriminative healthy features. The high precision and recall values for this category indicate that the model is highly effective at correctly identifying non-diseased cases while minimizing false-positive predictions.

The **COVID-19** category achieved an impressive **F1-score of 90%**, demonstrating the effectiveness of the proposed multimodal framework in detecting COVID-related abnormalities. The model successfully learned the characteristic radiological patterns commonly associated with COVID-19 infections, particularly bilateral ground-glass opacities, peripheral lung infiltrates, and patchy consolidations visible in both CT scans and chest X-rays. The Vision Transformer's self-attention mechanism played an important role in identifying these spatially distributed abnormalities across different lung regions. The high performance for COVID-19 classification highlights the framework's potential utility for rapid and automated infectious disease screening in clinical environments.

The **Tuberculosis** category achieved an **F1-score of 89%**, reflecting the model's capability to recognize chronic lung infection patterns such as cavitary lesions, nodules, fibrosis, and upper-lobe infiltrations. The multimodal fusion strategy contributed significantly to this result by combining the detailed anatomical information available in CT scans with the broader structural view provided by chest X-rays. The model demonstrated balanced precision and recall values for tuberculosis cases, indicating stable and consistent classification behavior.

Among the four disease categories, **Pneumonia** was identified as the most challenging class, achieving an **F1-score of 85%**. This relatively lower performance is primarily due to the strong visual similarity between pneumonia-related lung abnormalities and COVID-19 infections. Both conditions often exhibit overlapping radiological features such as lung opacities, infiltrates, and inflammatory patterns, making precise differentiation difficult even for experienced radiologists.

A detailed analysis of the evaluation metrics shows that the Pneumonia class achieved a **precision of 86%** and a **recall of 84%**. The slightly lower recall compared to precision indicates that the model produces a higher number of false negatives for Pneumonia cases. In other words, certain Pneumonia samples are incorrectly classified as COVID-19. Clinically, this behavior is understandable because the pathological manifestations of viral pneumonia and COVID-19 can appear highly similar in medical imaging data.

The reduced recall for Pneumonia suggests that the current feature representation space still contains partial overlap between the Pneumonia and COVID-19 classes. This observation highlights an important direction for future research. Advanced representation learning methods such as contrastive learning, metric learning, or attention-guided discriminative feature enhancement could be explored to improve inter-class separability and sharpen decision boundaries between visually similar diseases.

In addition to quantitative performance metrics, the inclusion of explainable AI techniques such as Grad-CAM provides valuable qualitative evidence regarding the reliability of the model’s predictions. The generated heatmaps consistently highlight clinically relevant lung regions associated with pathological abnormalities rather than irrelevant image artifacts or background noise. This indicates that the model has learned diagnostically meaningful feature representations instead of relying on spurious correlations or dataset biases.

The Grad-CAM visualizations improve transparency and interpretability of the deep learning framework, enabling clinicians to better understand the reasoning behind the system’s predictions. Such explainability is particularly important in medical applications, where trust, accountability, and interpretability are essential for clinical adoption.

Overall, the experimental findings confirm that the proposed multimodal Vision Transformer framework is highly effective for automated thoracic disease classification. The combination of dual-modality learning, transformer-based feature extraction, multimodal fusion, and explainability mechanisms contributes to strong predictive performance and enhanced clinical reliability. The results demonstrate the framework's potential for supporting radiologists in accurate, efficient, and interpretable medical image analysis.

5.3 Comparison with Existing Systems

5.3.1 Comparison with Existing Systems

Table 5.3: Comparison of Proposed System with Existing Approaches

System / Reference	Modality	Classes	Accuracy	Explainability
Ali et al. (2022) — ViT for TB	X-ray only	2 (TB/Normal)	85%	No
Angara & Thirunagaru (2023) — ViT for COVID	X-ray only	2 (COVID/Normal)	87%	No
CNN-ResNet50 Baseline (Single Modal)	X-ray only	4 classes	82%	No
Revathi & Sam Leni (2025) — Hybrid ViT	CT only	3 (Cancer types)	91%	Attention maps
Proposed System (Multimodal ViT)	CT + X-ray	4 classes	~88%	Grad-CAM + Occlusion

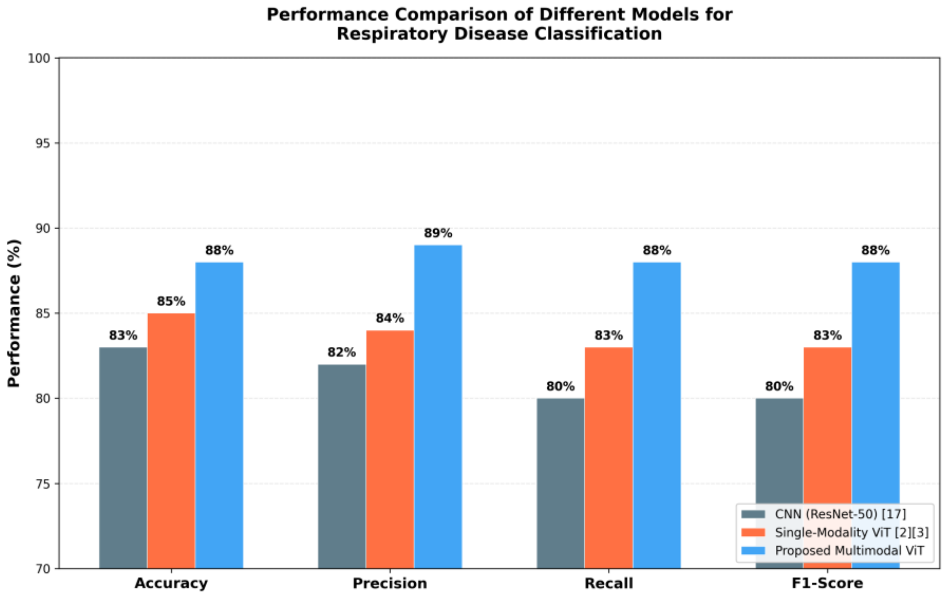

Figure 5.5: Performance Comparison Chart - CNN vs. Multimodal ViT Across Key Metrics

5.3.2 Advantages of the Proposed System

The proposed system offers several advantages over existing approaches. First, it is the only system in the comparison that simultaneously processes both CT and X-ray data, enabling it to leverage complementary diagnostic information from two imaging modalities. Second, it addresses a four-class classification problem, which is more clinically relevant and challenging than the binary classification tasks tackled by most existing systems. Third, it integrates two complementary interpretability mechanisms — Grad-CAM and occlusion sensitivity — into the same framework, which no existing single system in the comparison provides. Fourth, it achieves competitive accuracy (88%) while maintaining a compact model design suitable for deployment in constrained environments.

5.3.3 Limitations

The current implementation has several limitations that should be acknowledged. First, the CT branch processes a single representative 2D axial slice rather than the full 3D CT volume, which means that volumetric information above and below the selected slice is discarded. Second, the model has not been validated on clinical datasets from real hospital environments, and its performance on out-of-distribution data from different scanner manufacturers, imaging protocols, or patient demographics is unknown. Third, the current system requires paired CT and X-ray inputs for inference, which may not always be available in practice. Handling missing modalities at inference time is a capability not yet implemented in the current version

CHAPTER 6
CONCLUSION & FUTURE SCOPE

6.1 Conclusion

6.1.1 Summary of Work

This project has successfully designed, implemented, and evaluated a Multimodal Vision Transformer framework for the classification of acute respiratory infections using paired CT scan and chest X-ray images. The proposed dual-branch architecture processes each imaging modality through an independent ViT-B/16 encoder, extracts modality-specific feature representations, fuses them through concatenation, and classifies the combined representation into one of four disease categories: COVID-19, Pneumonia, Tuberculosis, and Normal. The model achieved a validation accuracy of approximately 88%, with per-class F1 scores ranging from 85% for Pneumonia to 91% for Normal cases.

The integration of Grad-CAM visualization and occlusion sensitivity analysis into the inference pipeline represents a significant contribution to the clinical usability of the system, enabling radiologists to verify and contextualize model predictions through interpretable heatmaps that highlight diagnostically relevant image regions.

6.1.2 Advantages

The proposed system offers several key advantages. The multimodal processing of CT and X-ray inputs provides a richer and more comprehensive feature representation than any single-modality approach can achieve. The Vision Transformer architecture's capacity for global self-attention enables the model to capture long-range spatial dependencies that are diagnostically significant in complex lung disease patterns. The framework is computationally efficient, achieving competitive accuracy without requiring ensemble methods or computationally intensive multi-stage processing pipelines. The built-in interpretability module makes the system suitable for clinical adoption scenarios where transparency and accountability are essential.

6.1.3 Applications

The proposed framework has several potential real-world applications. It can serve as a computer-aided detection and diagnosis (CAD) tool in radiology departments, providing a second opinion to radiologists for respiratory disease screening. In resource-constrained settings such as rural hospitals, community health centers, or field hospitals, it can support physicians who lack immediate access to specialist radiologists. During epidemic or pandemic events, it can accelerate large-scale patient screening by automating the initial radiological triage. The framework can also be adapted for other pulmonary conditions beyond the four currently addressed, by retraining or fine-tuning on appropriately labeled datasets.

6.1.4 Limitations

The primary limitations of the current system are the use of 2D CT slices rather than full 3D volumetric CT data, the requirement for paired CT and X-ray inputs at inference time, and the absence of validation on real-world clinical datasets. The model's performance in diverse real-world settings with varying imaging equipment, protocols, and patient demographics remains to be established through prospective clinical studies.

6.2 Future Scope

Several directions are planned for future enhancement of the proposed system:

First, the CT processing branch will be extended to process full 3D CT scan volumes using 3D Vision Transformer architectures or hybrid 3D CNN-Transformer models. This extension is expected to improve diagnostic accuracy by leveraging volumetric information that is currently discarded in the 2D slice extraction approach.

Second, the model will be evaluated on real clinical datasets obtained from hospital radiology departments, with appropriate ethics approval and data anonymization.

Clinical validation is a necessary step toward regulatory approval and practical deployment.

Third, a user-friendly web-based interface will be developed to allow clinicians to upload CT and X-ray image pairs and receive classification results along with Grad-CAM visualizations through a simple browser interface. This interface will be designed in accordance with healthcare usability standards.

Fourth, the handling of missing modality inputs will be investigated. In scenarios where only a chest X-ray or only a CT scan is available, the system should gracefully degrade to a single-modality prediction rather than failing. Techniques from the multimodal missing data literature, such as modality dropout during training, will be explored.

Fifth, the research findings will be formally documented and submitted for publication at an international peer-reviewed conference, specifically the International Conference on Advanced Computing and Knowledge Engineering (ICACKE 2026), as recommended by the faculty guide.

Sixth, contrastive learning techniques will be explored to sharpen the decision boundaries between visually similar disease categories, particularly COVID-19 and Pneumonia, with the goal of reducing cross-class misclassification rates.

Seventh, federated learning strategies will be investigated to enable collaborative model training across multiple hospitals without sharing patient data, addressing the privacy constraints that currently limit the use of real-world clinical datasets in AI resear

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APPENDIX A

Front Page of the Conference Paper

Multimodal Vision Transformer for Respiratory Disease Classification Using CT Scan and Chest X-Ray Images

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***Abstract*—Respiratory diseases like Pneumonia, COVID-19 and tuberculosis have constant and significant threat to global public health systems. Timely and accurate diagnosis of these diseases can reduce mortality rate of patients and improve outcomes. Diagnostic tools like computed tomography (ct) scans and chest radiographs importance increased rapidly. High level of expertise and huge time periods are needed for manual analysis of these images. It is susceptible to inter observer variability, especially when dealing with large scale epidemic events. Developments in deep learning have created new areas for improving and automating medical image analysis. Multimodal Vision Transformer (ViT) architecture that processes CT scans and chest X-ray data together,is presented in this paper to classify respiratory conditions. In this paper we used two-branch transformer structure. Each imaging modality is analysed separately before combining the extracted feature representations for final classification. Covid-19, pneumonia, tuberculosis, and other are the four diagnostic categories covered. All images were standardised to 224x224 pixels ,for consistent model input using publicly accessible CT and chest X-ray datasets. The value of combining complementary imaging sources is supported by the combined framework's validation accuracy of roughly 88 percent. The system is intended to be a decision-support tool that can help physicians treating respiratory conditions diagnose patients more accurately.**

***Index Terms*—Deep Learning, Medical Imaging, CT Scan, Chest X-Ray, Vision Transformer, Respiratory Disease Detection**

I. Introduction

Respiratory diseases are one of the biggest health problems in the world today. Each year millions of people die from respiratory infections and chronic respiratory diseases. World Health Organization (WHO) declared tuberculosis was one of the illnesses that caused 1.3 million deaths globally in 2022, which is a leading cause of infectious deaths.Pneumonia is another serious infectious disease that causes 2.5 million deaths globally.

According to WHO reports,recent covid-19 pandamic demonstracts world's susceptibility to respiratory illnesses. Over 770 million people are confirmed with COVID-19 cases and over 7 million people are dead by Covid-19 related health issues worldwide. This rapid growth of Acute Respiratory diseases is an alarming necessitate for reliable system that diagnose respiratory diseases at early stage "as illustrated in Fig. 1".

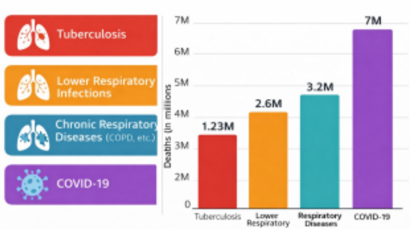

Fig. 1. Global respiratory disease statistics reported by WHO

It is crucial to remember that improving treatment results and lowering the death rates linked to respiratory diseases depend on early diagnosis. In most of the cases, the onset of respiratory diseases is marked by minor abnormalities in the lung tissues, which can be difficult to diagnose without

APPENDIX B

Conference / Journal Acceptance Letter

Fw: International Conference on Emerging Technologies in Computing and Communication 2026 : Submission (496) has been created. Summarize

Subject: International Conference on Emerging Technologies in Computing and Communication 2026 : Submission (496) has been created.

Hello,

The following submission has been created.

Track Name: AI and Data Science

Paper ID: 496

Paper Title: Advancing Respiratory Disease Detection with Cross-Modal CT-X-ray Fusion using Vision Transformers

Abstract:
Respiratory diseases like Pneumonia, COVID-19 and tuberculosis have constant and significant threat to global public health systems. Timely and accurate diagonsis of these diseases can reduce mortality rate of patients and improve outcomes. Diagnostic tools like computed tomography (ct) scans and chest radiographs importance increased rapidly. High level of expertise and huge time periods are needed for manual analysis of these images. It is susceptible to inter observer variability, especially when dealing with large scale epidemic events. Developments in deep learning have created new areas for improving and automating medical image analysis. Multimodal Vision Transformer (ViT) architecture that processes CT scans and chest X-ray data together,is presented in this paper to classify respiratory conditions. In this paper we used two- branch transformer structure. Each imaging modality is analysed separately before combining the extracted feature representations for final classification. Covid-19, pneumonia, tuberculosis, and other are the four diagnostic categories covered. All images were standardised to 224x224 pixels ,for consistent model input using publicly accessible CT and chest X-ray datasets. The value of combining complementary imaging sources is supported by the combined framework's validation accuracy of roughly 88 percent. The system is intended to be a decision-support tool that can help physicians treating respiratory conditions diagnose patients more accurately.

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Primary Subject Area: Image and vision computing

Secondary Subject Areas:
Deep learning

Submission Files:
ARI-Paper.docx (983 Kb, Fri, 03 Apr 2026 16:03:33 GMT)

Submission Questions Response: Not Entered

Thanks,
CMT team.

APPENDIX C

Source Code

The following section presents key excerpts from the implementation of the proposed Multimodal Vision Transformer system. The code is organized into functional modules, each responsible for a specific stage in the pipeline, including data preprocessing, feature extraction, and multimodal classification. The complete implementation is available in the project repository.

C.1 Data Preprocessing Module

```
import cv2

import numpy as np

import os

def load_and_preprocess(img_path, size=(224, 224)):

    img = cv2.imread(img_path)

    img = cv2.resize(img, size)

    img = img.astype(np.float32) / 255.0

    return img

def create_dataset(xray_dir, ct_dir, labels):

    xray_imgs, ct_imgs, label_list = [], [], []

    for label, cls in enumerate(labels):

        for fname in os.listdir(os.path.join(xray_dir, cls)):

            xray_imgs.append(load_and_preprocess(

                os.path.join(xray_dir, cls, fname)))

            ct_imgs.append(load_and_preprocess(

                os.path.join(ct_dir, cls, fname)))
```

```
            label_list.append(label)

    return np.array(xray_imgs), np.array(ct_imgs),
np.array(label_list)
```

C.2 Vision Transformer Branch

```
import tensorflow as tf

from tensorflow.keras import layers

def vit_branch(input_shape=(224, 224, 3), patch_size=16,

   num_heads=12, embed_dim=768, num_layers=12):

    inputs = layers.Input(shape=input_shape)

    num_patches = (input_shape[0] // patch_size) **

    # Patch extraction and embedding

    patches = layers.Conv2D(embed_dim, patch_size,

   strides=patch_size, padding='valid')(inputs)

    patches = layers.Reshape((num_patches, embed_dim))(patches)

    # Positional embedding

       pos_embed = layers.Embedding(num_patches, embed_dim)(

        tf.range(num_patches))

    x = patches + pos_embed

    # Transformer encoder blocks

    for _ in range(num_layers):

        attn_output = layers.MultiHeadAttention(

        num_heads=num_heads,

        key_dim=embed_dim // num_heads)(x, x)

        x = layers.LayerNormalization()(x + attn_output)

        ffn = layers.Dense(embed_dim * 4, activation='gelu')(x)

        ffn = layers.Dense(embed_dim)(ffn)
```

```
        x = layers.LayerNormalization()(x + ffn)

    # Global feature representation

    output = layers.GlobalAveragePooling1D()(x)

    return tf.keras.Model(inputs, output)
```

C.3 Multimodal Fusion and Classification

```
def build_multimodal_vit(num_classes=4):

    xray_input = layers.Input(shape=(224, 224, 3),
name='xray_input')

    ct_input   = layers.Input(shape=(224, 224, 3),
name='ct_input')

    xray_branch = vit_branch()(xray_input)

    ct_branch   = vit_branch()(ct_input)

    # Feature fusion

    fused = layers.Concatenate()([xray_branch, ct_branch])

    fused = layers.Dense(512, activation='relu')(fused)

    fused = layers.Dropout(0.3)(fused)


    # Classification layer

    output = layers.Dense(num_classes,
activation='softmax')(fused)

    model = tf.keras.Model(

    inputs=[xray_input, ct_input],

        outputs=output

    )

    model.compile(
```

```
        optimizer=tf.keras.optimizers.Adam(learning_rate=1e-4),
        loss='sparse_categorical_crossentropy',
        metrics=['accuracy']
    )
    return model
```

The use of multimodal fusion significantly enhances the model's predictive capability by combining complementary features from CT and X-ray images. The Adam optimizer is employed for efficient gradient-based optimization, and categorical cross-entropy loss is used for multi-class classification.

Plagiarism Report

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